

Seeing the Invisible

Physics-Guided Missing-Physics Learning for PEMFC Flooding Prediction

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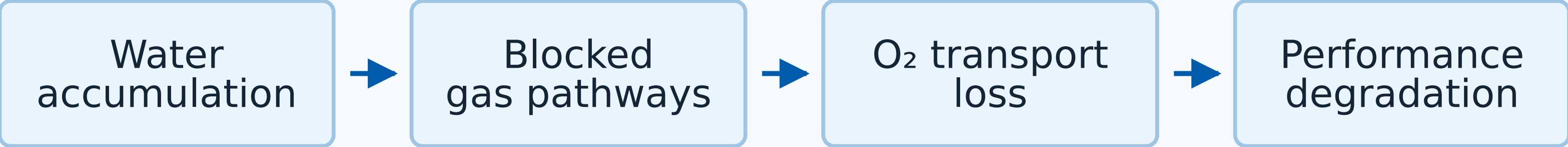


ABSTRACT

Liquid water accumulation in porous gas diffusion layers critically limits mass transport in polymer electrolyte membrane fuel cells, yet the internal flooding state is difficult to observe directly. More broadly, physics-based multiphysics models often rely on empirical or fitted constitutive relations to represent unresolved transport behavior, making constitutive uncertainty a major source of prediction error. Here, we propose a physics-informed grey-box framework that retains the known governing equations within a finite-element solver while learning only the missing constitutive physics. A learnable gas-phase transport closure is embedded directly into the two-phase FEM model and identified from synthetic reference responses using discrete-adjoint sensitivities, without prescribing a fixed functional form in advance. After training at RH 50%, the learned constitutive law is frozen and evaluated at unseen RH 70% on two discretizations, 16×40 and 32×80 . Agreement with the synthetic reference is retained without retraining in the reported test cases, supporting transfer of the learned closure across the tested humidity and discretization changes. These synthetic tests motivate future experimental validation for flooding diagnosis and physics-guided digital twins.

INTRODUCTION

Why flooding matters



What limits flooding prediction?

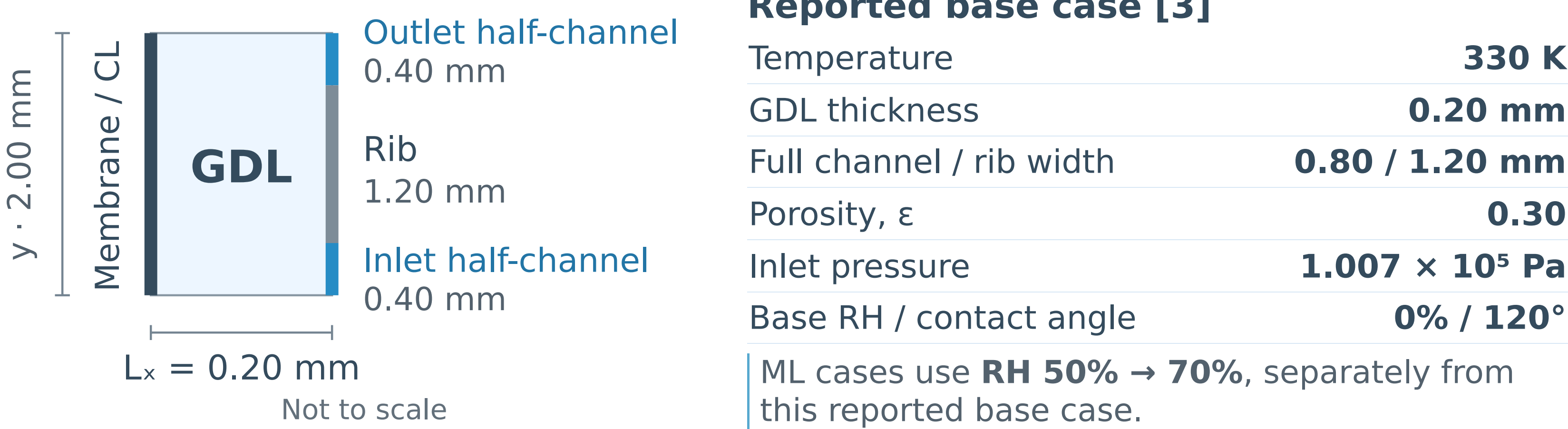
Observation	Internal liquid-water saturation $s(x,y)$ cannot be directly and continuously observed during operation.
Modeling	Flooding predictions depend on uncertain constitutive relations such as $D_{\text{eff}}(s)$, $k_r(s)$, and $p_c(s)$.

Existing ML vs our learning objective

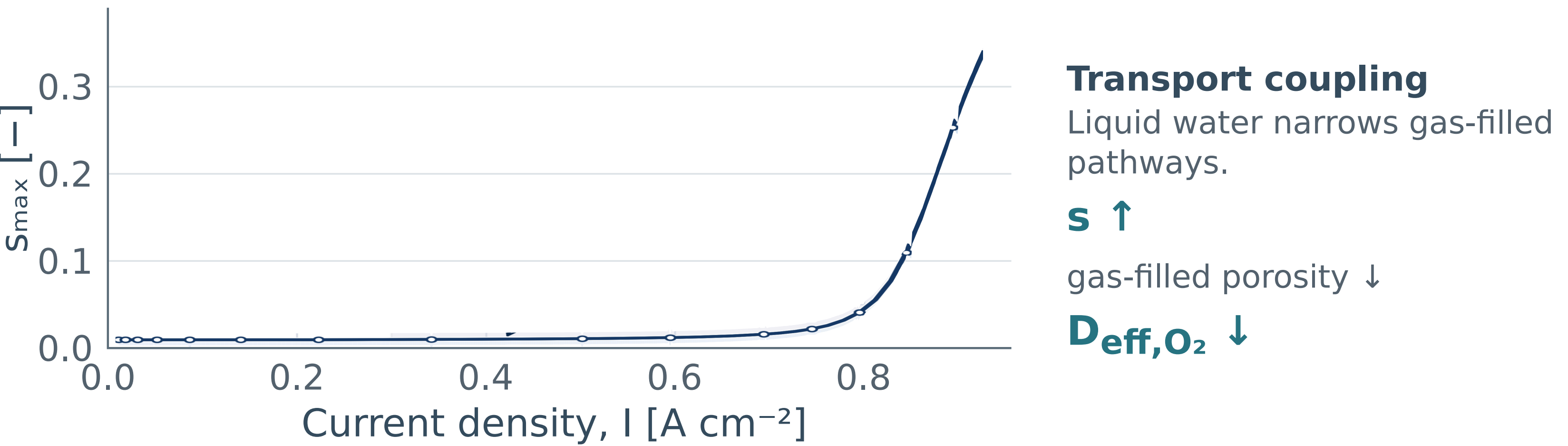
PINN Learn the PDE solution Physics is enforced through equation-residual training.	FNO / Neural operator Learn the solution operator Map input fields or parameters to solution fields.	OUR APPROACH FEM + Missing Physics ML Learn the constitutive law Known conservation laws remain in FEM; only the uncertain transport relation is learned.
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RESULTS

01 Model domain & reference conditions



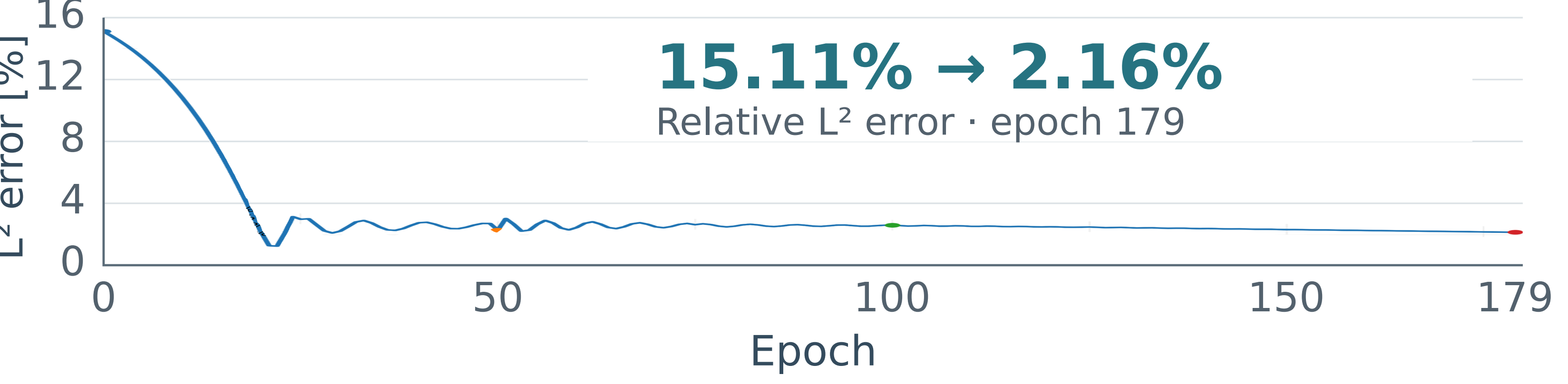
02 Water accumulation grows at high current



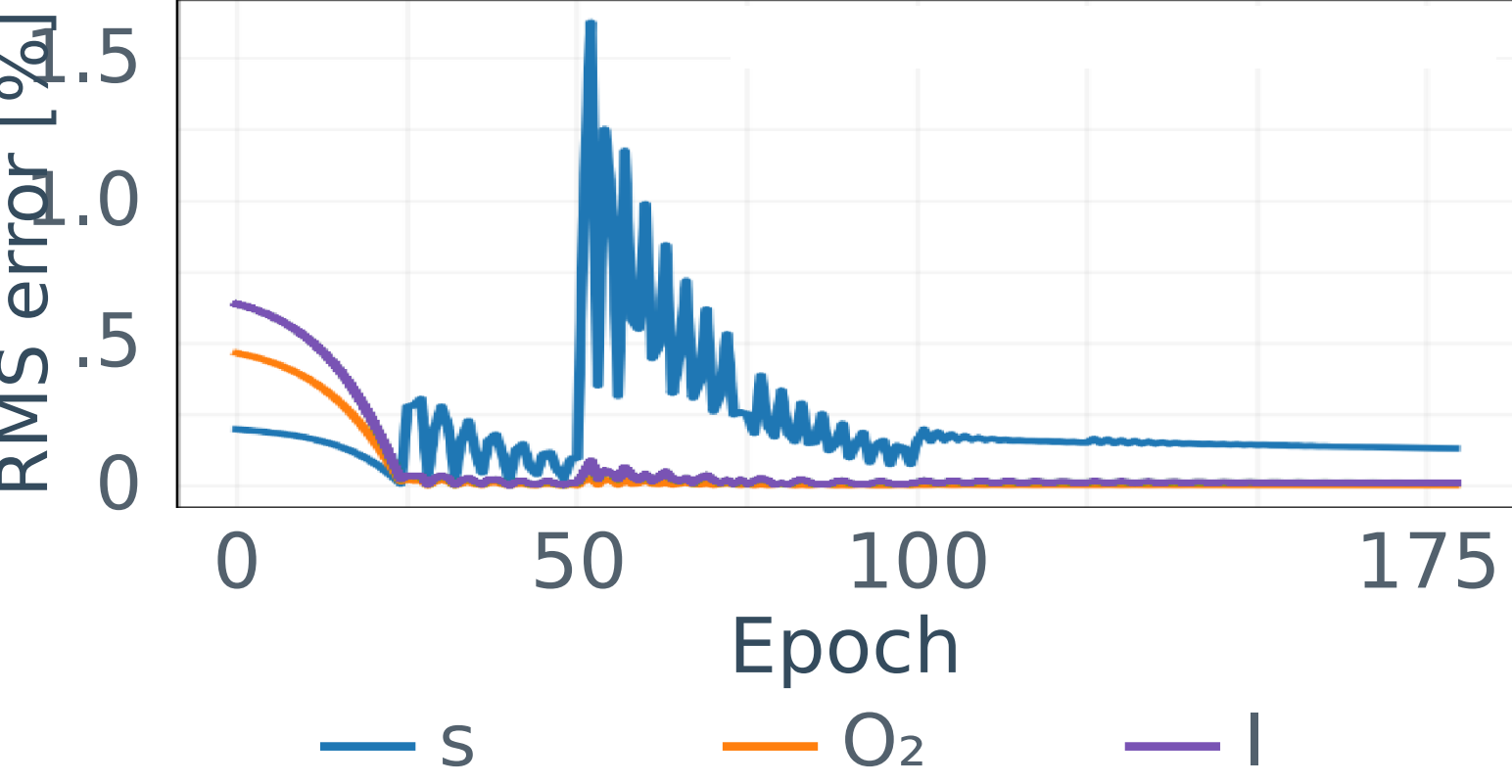
S_{max} : maximum liquid saturation within the GDL.

03 Recover the missing transport relation

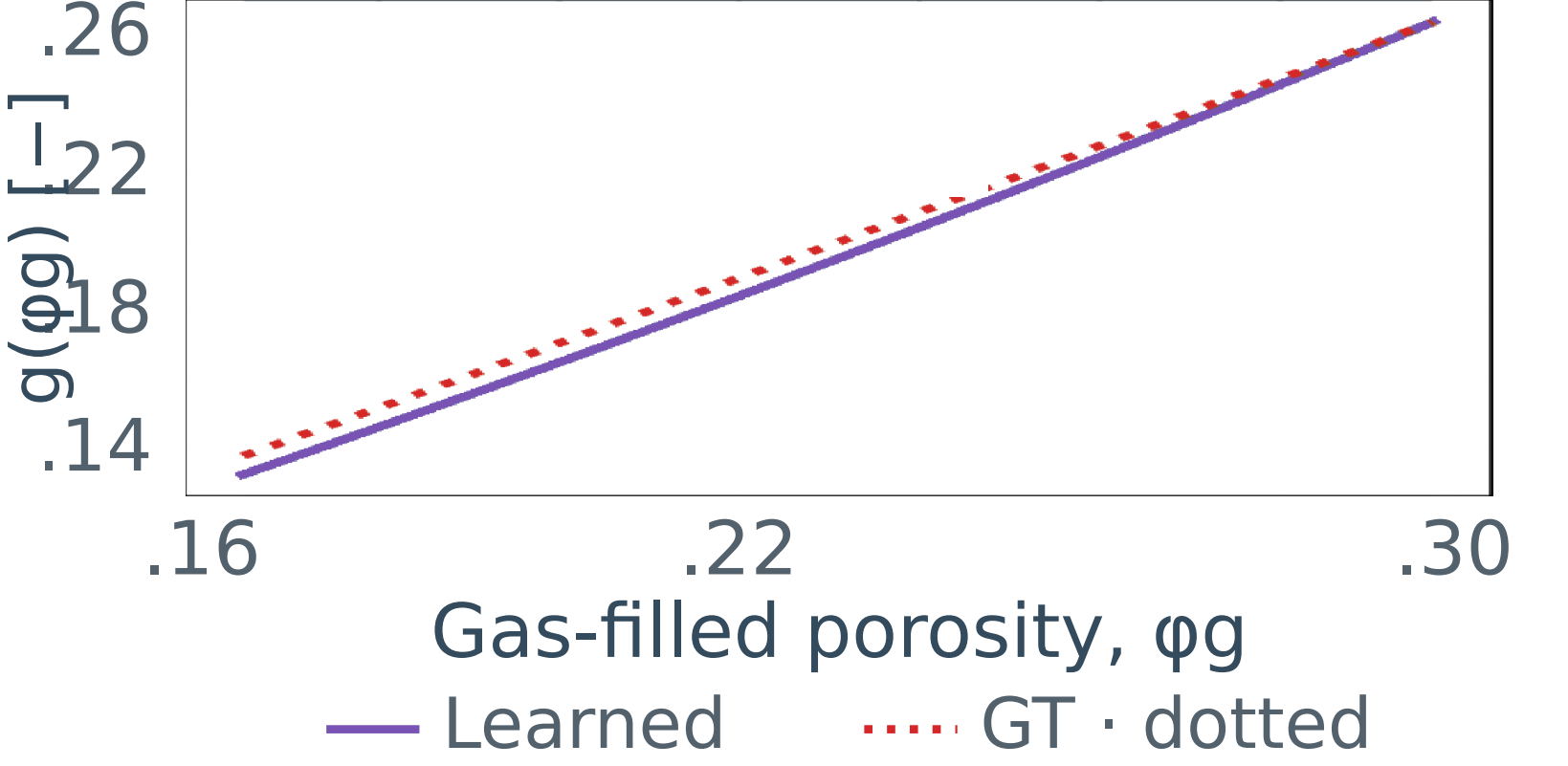
Train at **RH 50%**: infer the gas-phase closure from synthetic FEM responses.



State/response reconstruction

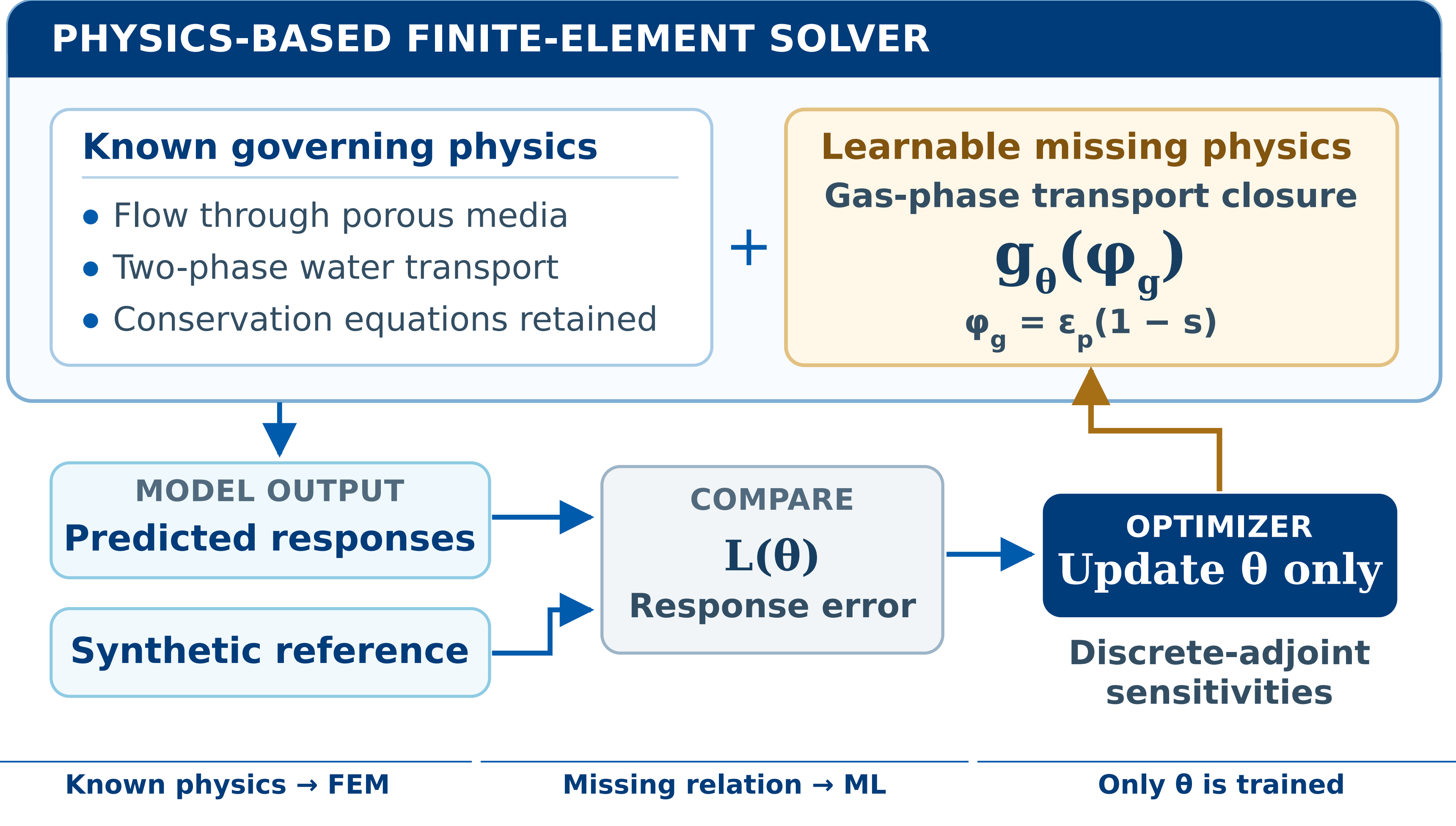


Recovered constitutive law

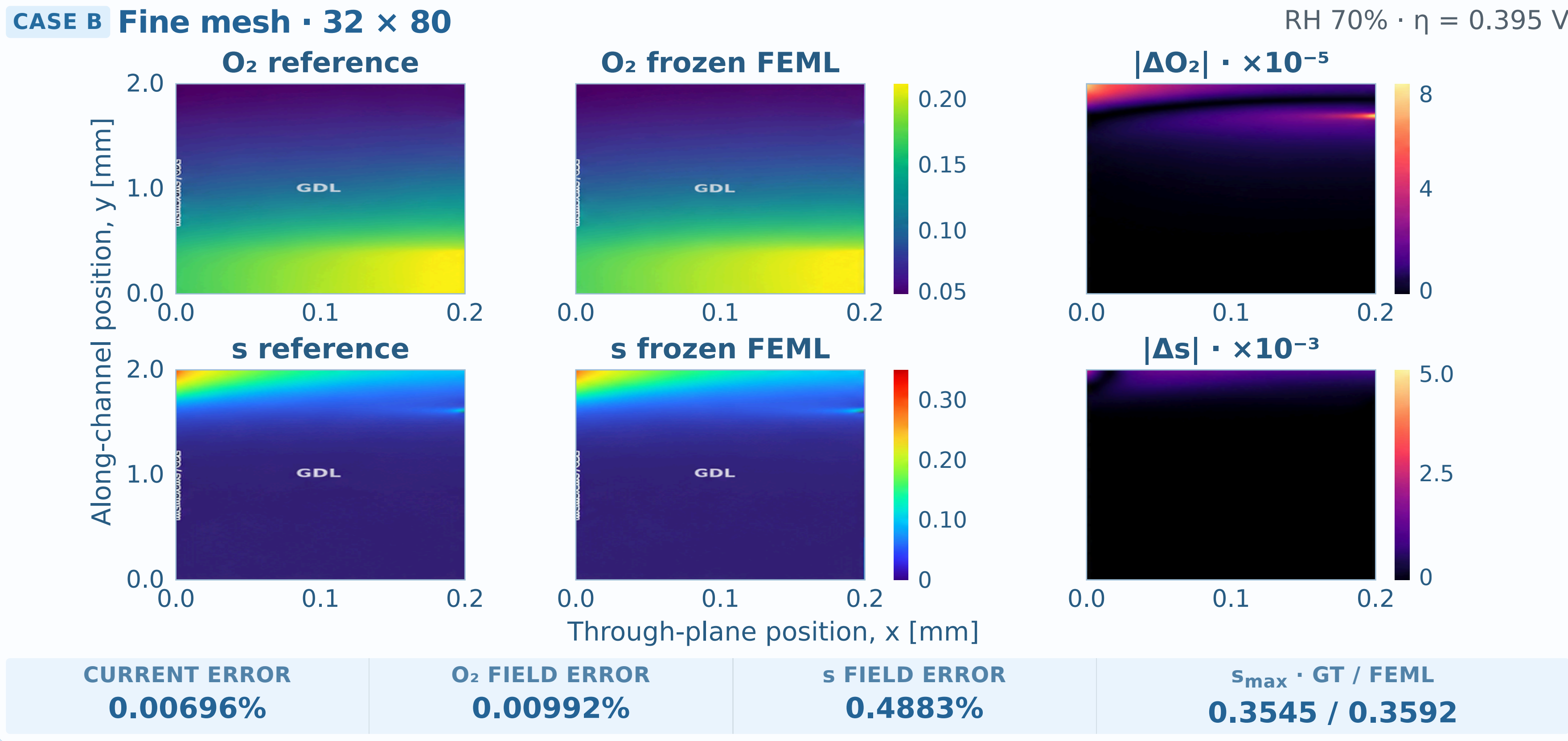
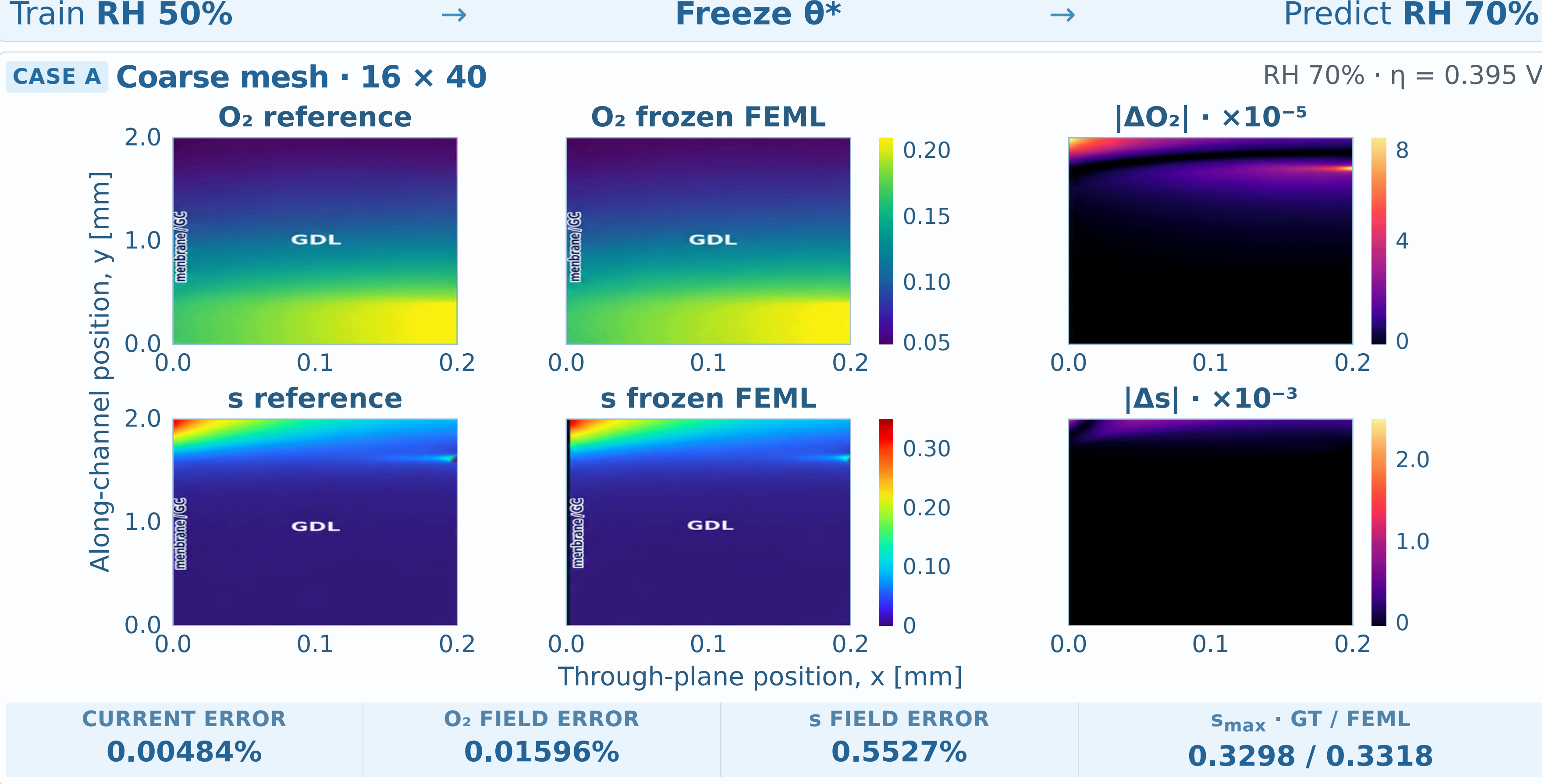


$D_{\text{eff}} = D_{\text{dry}} g_{\theta}(\phi_g)$ $\phi_g = \epsilon_p(1 - s)$ GT: $g(\phi_g) = \phi_g^{\tau}$

MODELING FRAMEWORK



04 Zero-shot transfer with a frozen closure



Shared reference/prediction scales; absolute error maps. x is stretched for display.

Conclusions & outlook

A learned transport law can be reused inside FEM without retraining.

01 Physically interpretable learning. The recovered closure links gas-filled porosity to transport, making the learned relation directly inspectable.

02 Transfer within the tested conditions. Agreement at changed RH and discretization supports reuse of the closure; it does not establish mesh convergence.

03 Next step: predictive validation. Test against experimental responses and a wider operating range, then assess closure identifiability and uncertainty.